

GenHack 2025 Urban Heat Island Analysis

Berlin Case Study Using ERA5-Land, Sentinel-2 NDVI, GADM Boundaries, and ECA&D Ground Stations

Portfolio Analysis Report | Python / GeoPandas / Rasterio / Xarray

Prepared as a professional personal-project report based on Team 46 repository materials and the GenHack 2025 participant brief.

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Executive Summary. This project studies the Urban Heat Island (UHI) problem by integrating multiple geospatial and climate datasets. The analysis focuses on Berlin because the repository contains a clean Berlin boundary extracted from GADM, 20 ECA&D weather stations inside the city, and 2020-2023 Sentinel-2 NDVI observations that reveal meaningful vegetation variation across the urban area. The workflow transforms satellite vegetation rasters, ERA5-Land gridded meteorological data, and ground-station temperature observations into comparable spatial and temporal formats. The result is a reproducible Python analysis pipeline that can support UHI measurement, visualization, and future explanatory modeling.

1. Challenge Context and Project Goal

GenHack 2025 was a four-week collaborative hackathon on Urban Heat Islands and Climate Downscaling. The participant brief asked teams to explore the UHI effect using ERA5 temperature data, Sentinel-2 NDVI maps, and ground-station observations. The evaluation criteria emphasized data integration, visualization and communication, metrics and quantitative insight, explanatory modeling, and collaboration. This report converts the team repository into a complete data-analysis story suitable for internship or research-assistant portfolios.

The core analytical question is: how can we combine satellite vegetation, reanalysis climate grids, administrative boundaries, and station observations to describe local heat patterns in an urban area? A successful project must do more than make maps. It needs to define the spatial unit, clean and align heterogeneous data sources, construct interpretable metrics, and communicate what the results imply about urban climate exposure.

Project deliverables reconstructed in this report

- A city-level UHI problem definition focused on Berlin.
- A reproducible Python workflow for GADM boundary extraction, NDVI raster masking, ERA5 grid handling, and ECA&D station filtering.
- A measurement strategy for UHI using vegetation, gridded temperature, and station observations.
- Visual outputs and interpretations based on the repository notebooks and saved figures.
- A limitations and extension section that explains what would be needed to turn the exploratory work into a stronger production-style model.

2. Why Berlin Was a Strong Study Area

The repository contains both early Barcelona exploration and a Berlin-focused workflow. For the final portfolio framing, Berlin is the stronger case study. Berlin is a major European city with dense built

surfaces, large parks and forests, waterways, and enough internal land-cover variation to make UHI analysis meaningful. It also has strong data support in the available files.

- Clean boundary extraction: in the main Berlin notebook, filtering GADM with country code DEU and NAME_2 == Berlin returned exactly one city-level record. This avoids ambiguity from nested administrative units.
- Ground-station coverage: the ECA&D station workflow found 20 registered weather stations within Berlin. This supports a station-based validation path rather than relying only on gridded data.
- Visible vegetation contrast: the Sentinel-2 NDVI map shows strong spatial differences between dense urban surfaces, parks, forests, and water-adjacent areas.
- Good fit with GenHack goals: Berlin allows the project to connect administrative boundaries, satellite NDVI, ERA5-Land grids, and station observations in one city-level workflow.

```
country_code = "DEU" # Germany
cityname = "Berlin"

filtered_gadm_gdf = gadm_gdf[(gadm_gdf.GID_0 == country_code) &
                              (gadm_gdf.NAME_2 == cityname)]
print(f"Number of rows in GADM after filtering: {len(filtered_gadm_gdf)}")

one_city_gdf = filtered_gadm_gdf.dissolve()
```

Repository output: the Berlin filter returned one row, and the dissolved geometry was used as the city boundary for subsequent raster masking and station filtering.

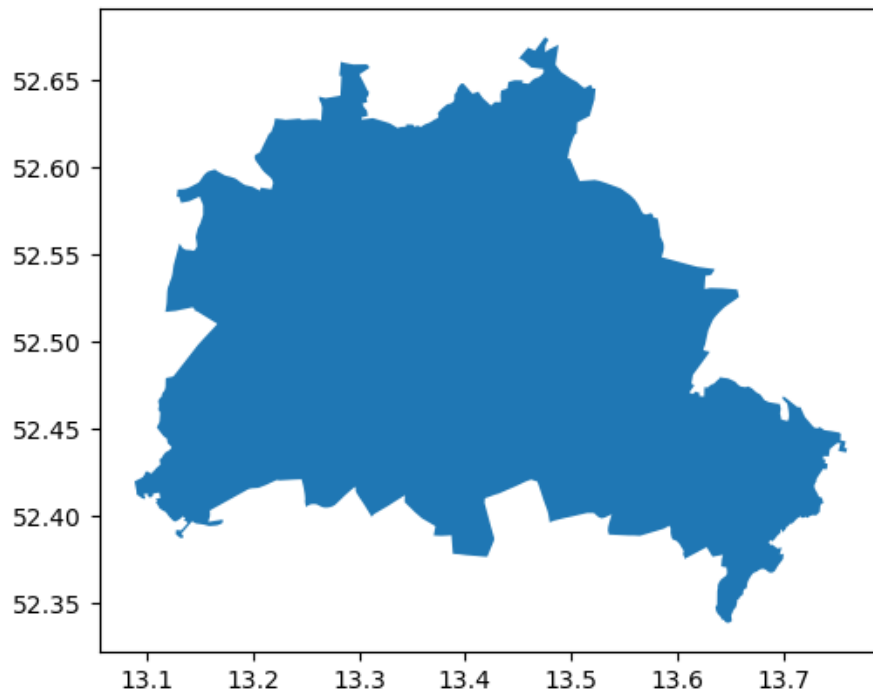


Figure 1. Berlin city boundary extracted from GADM and used as the common spatial unit.

3. Data Sources and What Each Dataset Contributes

The project is valuable because it uses several datasets that each represent a different view of the same urban climate problem. UHI cannot be measured well from a single source. Satellite NDVI describes

surface greenness, ERA5-Land provides gridded meteorological context, ECA&D stations provide observed ground-level maximum temperature, and GADM provides administrative boundaries for spatial clipping.

Dataset	Repository evidence	Role in the analysis
GADM 4.1 Europe	106,252 administrative records; Berlin extracted as one record	Defines the city boundary and makes spatial masking possible.
Sentinel-2 NDVI	Quarterly GeoTIFF rasters for 2020-2023; stored on 0-254 scale with nodata = 255	Measures vegetation and land-cover variation, a key explanatory factor for urban heat.
ERA5-Land	Daily gridded climate variables stored as NetCDF; xarray workflow with 350 x 660 grid	Provides spatially continuous climate context such as temperature, wind, and precipitation.
ECA&D TX stations	8,568 total stations; 20 inside Berlin; selected station 4563 with 54,475 valid daily records	Provides observed daily maximum temperature for validation and UHI signal exploration.

GADM Europe Dataset - Exploratory Data Analysis Summary

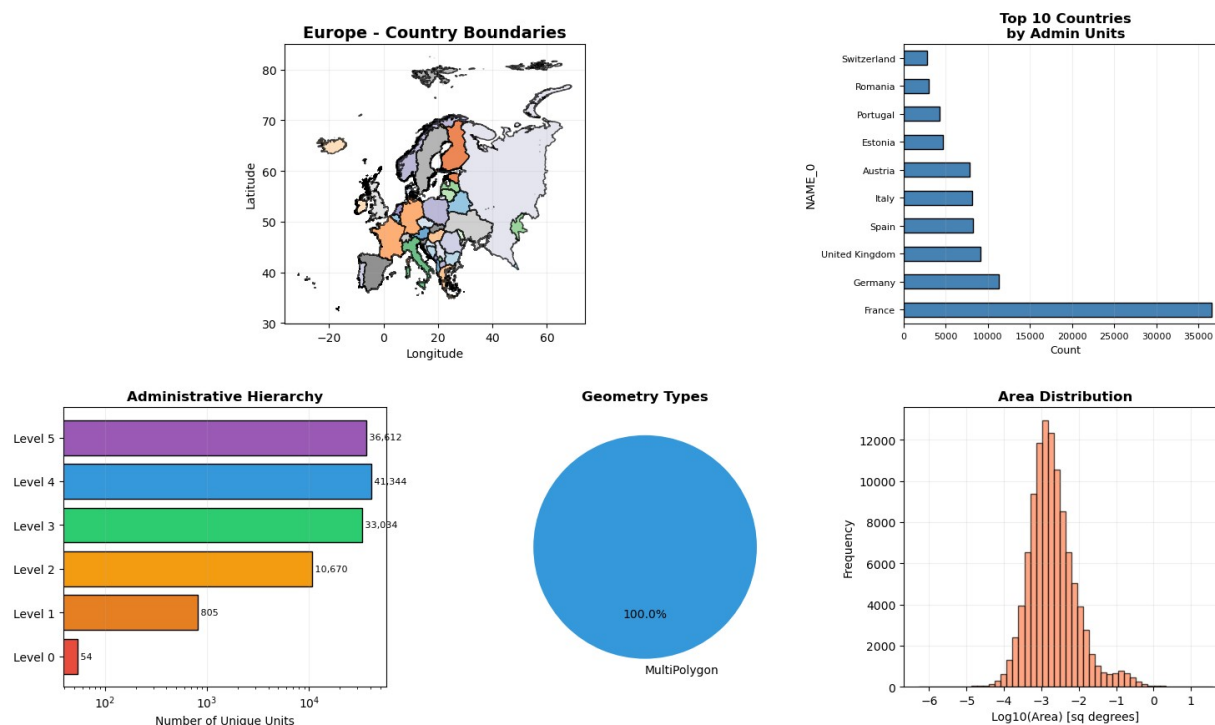


Figure 2. GADM Europe exploratory summary from the repository, showing administrative coverage and hierarchy.

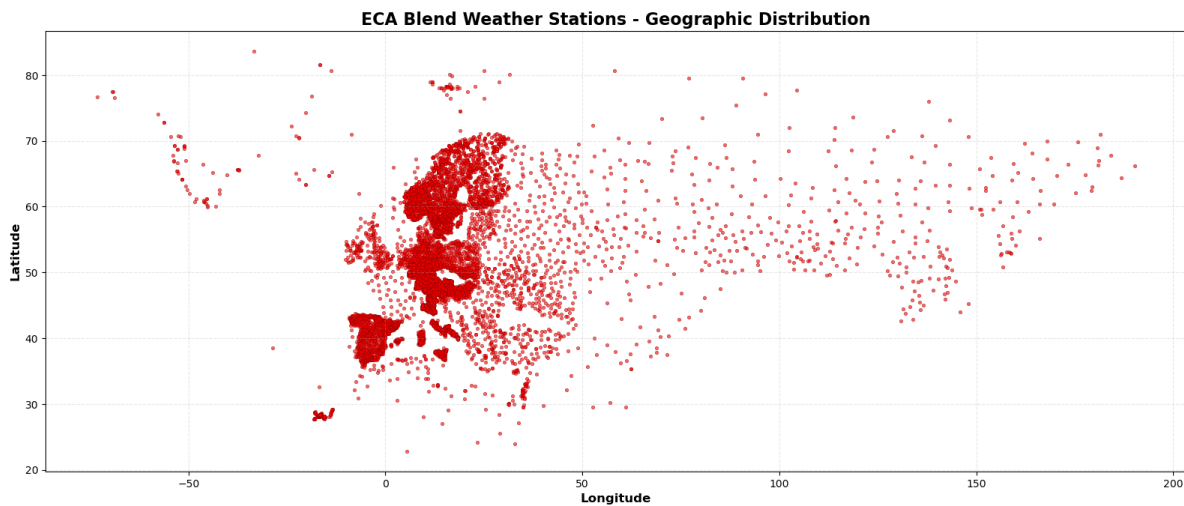


Figure 3. ECA&D station distribution across Europe, used to understand station coverage before city filtering.

4. End-to-End Workflow

The project workflow can be framed as an end-to-end data-integration pipeline. Each stage solves a specific problem: define the spatial unit, standardize data formats, align coordinate systems, compute interpretable metrics, and produce visuals that explain the city-level heat context.

1. Load GADM and extract the Berlin city boundary.
2. Load Sentinel-2 NDVI rasters and convert stored integer values into real NDVI values from -1 to 1.
3. Mask the NDVI raster to Berlin so only within-city pixels remain.
4. Compute quarterly mean NDVI and visualize the time series from 2020 to 2023.
5. Load ERA5-Land NetCDF variables using xarray and examine gridded meteorological patterns.
6. Reproject ERA5-Land variables to the NDVI grid so climate and vegetation layers become spatially comparable.
7. Load ECA&D station metadata, convert DMS coordinates to decimal degrees, and identify stations inside Berlin.
8. Load station-level daily maximum temperature, keep valid observations, and convert 0.1 degree Celsius units into Celsius.
9. Use the integrated data to propose UHI metrics and future model improvements.

4.1 NDVI processing and scaling

Sentinel-2 NDVI values in the repository are stored as integers. Before interpretation, the notebook converts them to the standard -1 to 1 NDVI scale and treats nodata values as missing. This is important because an unconverted raster would produce visually plausible maps but invalid numerical summaries.

```
def convert_ndvi_to_real_scale(ndvi_img, out_meta):
    # NDVI is stored on a 0/254 scale; nodata is 255.
    ndvi_img = ndvi_img.astype(float)
    ndvi_img[ndvi_img == out_meta["nodata"]] = np.nan
    ndvi_img = ndvi_img / 254 * 2 - 1
    return ndvi_img

def get_out_image_and_metadata(filepath, one_city_gdf):
    with rasterio.open(filepath) as src:
        one_city_gdf_in_good_crs = one_city_gdf.to_crs(src.crs)
```

```

city_geometry = [one_city_gdf_in_good_crs.geometry.iloc[0]]
out_image, out_transform = mask(src, city_geometry, crop=True)
out_meta = src.meta
out_meta.update({"height": out_image.shape[1],
                 "width": out_image.shape[2],
                 "transform": out_transform})
real_out_image = convert_ndvi_to_real_scale(out_image, out_meta)
return real_out_image, out_meta

```

This code shows the most important technical skill in the project: raw raster data were not used directly. They were transformed, masked, and converted into an interpretable form.

4.2 ERA5-Land handling and grid alignment

ERA5-Land files are NetCDF datasets. The repository uses xarray to open single-year and multi-year datasets, then uses rioxarray to reproject gridded variables onto the NDVI raster grid. This step matters because raster and climate grids often have different resolution, coordinate orientation, and transform metadata.

```

ds = xr.open_mfdataset(filepaths, combine="by_coords")
da = ds[datavar].sel(valid_time=day)

origin_transform = from_bounds(
    ds.longitude.min().item(), ds.latitude.min().item(),
    ds.longitude.max().item(), ds.latitude.max().item(),
    len(ds.longitude), len(ds.latitude)
)

reprojected_da = da.rio.write_crs("EPSG:4326").rio.reproject(
    dst_crs=out_meta["crs"],
    shape=(out_meta["height"], out_meta["width"]),
    transform=out_meta["transform"],
)

```

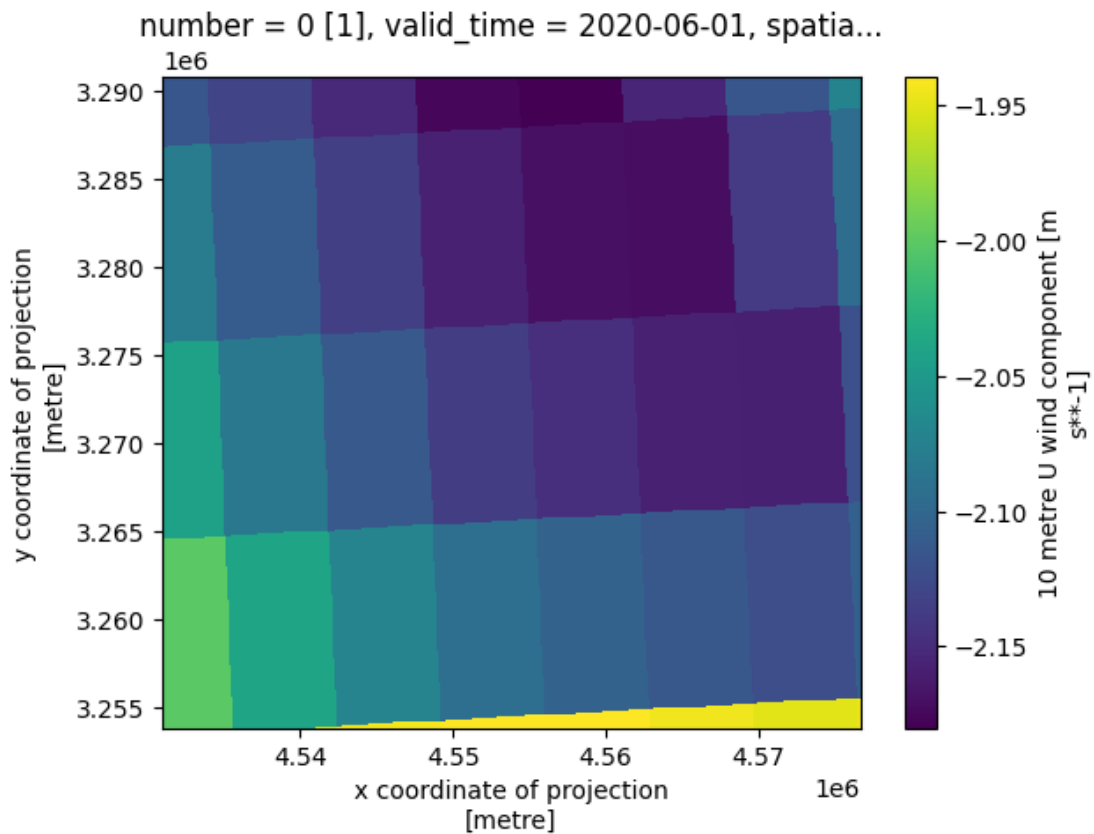


Figure 4. ERA5-Land variable reprojected to the Berlin NDVI grid, supporting later climate-vegetation comparison.

4.3 Ground-station data quality workflow

The ECA&D station files use DMS coordinates and station-level time-series files. The repository converts coordinates into decimal degrees, filters stations inside Berlin, and keeps only valid temperature observations using the Q_TX quality flag. This is a strong research-assistant style workflow because it documents data quality instead of treating all observations as automatically reliable.

```
def dms_to_decimal(dms_str):
    dms_str = dms_str.strip()
    sign = 1 if dms_str[0] == '+' else -1
    degrees, minutes, seconds = map(float, dms_str[1:].split(':'))
    return sign * (degrees + minutes/60 + seconds/3600)

stations_gdf = gpd.GeoDataFrame(
    stations_df,
    geometry=gpd.points_from_xy(stations_df['LON_decimal'],
                                stations_df['LAT_decimal']),
    crs="EPSG:4326"
)

stations_in_city_gdf = stations_gdf[stations_gdf.within(one_city_gdf.geometry.iloc[0])]
print(f"Number of registered stations within Berlin: {len(stations_in_city_gdf)}")
```

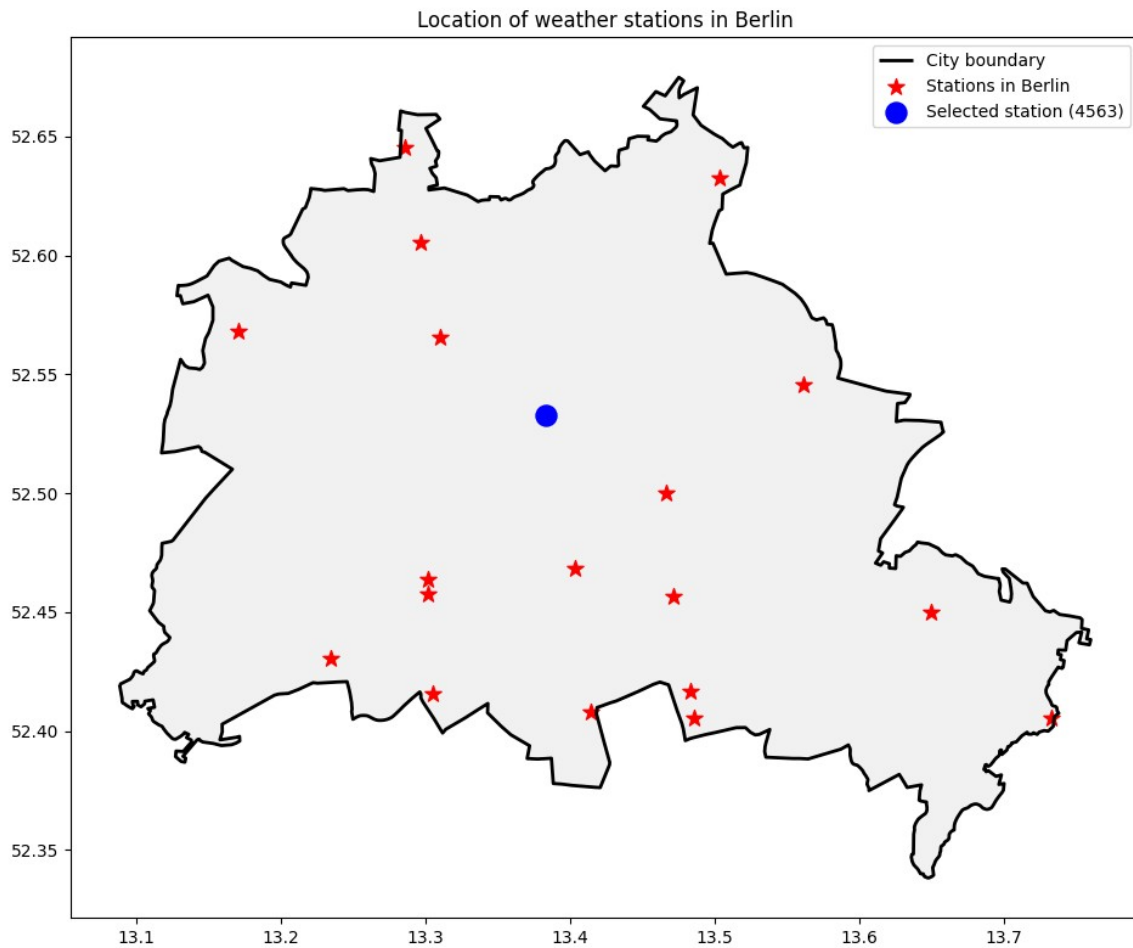


Figure 5. Twenty ECA&D weather stations were located inside the Berlin boundary; station 4563 was selected for time-series exploration.

5. How the Project Measures UHI

The Urban Heat Island effect is usually measured as a temperature difference between an urban area and a non-urban reference area. In a full production workflow, UHI intensity could be written as:

$$\text{UHI_intensity}(t) = T_{\text{urban}}(t) - T_{\text{reference}}(t)$$

The challenge is deciding what counts as urban and what counts as the reference. This project builds the foundation for that decision using three complementary sources:

- Station-based signal: use ECA&D daily maximum temperature at stations inside Berlin; later compare inner-city stations with peri-urban or rural reference stations.
- Gridded climate surface: use ERA5-Land to create a continuous climate field over the city and compare it with station observations.
- Vegetation proxy: use Sentinel-2 NDVI to classify pixels or areas by vegetation intensity; lower NDVI typically indicates built-up or bare surfaces, while higher NDVI indicates vegetated areas that may reduce local heat.

Because the repository mainly contains exploratory notebooks and saved outputs, the strongest completed metric is the vegetation time series and the station-temperature exploration. The next modeling step would compute temperature differences by station class or NDVI-defined zone and test

whether lower-NDVI areas systematically exhibit higher maximum temperature or stronger heat anomalies.

6. Results and Interpretation

6.1 Vegetation structure from Sentinel-2 NDVI

The NDVI map shows Berlin as a spatially heterogeneous city. Darker green areas indicate higher vegetation intensity, while lighter areas suggest lower vegetation or built-up surfaces. This pattern matters because UHI risk is not evenly distributed. Parks, forests, and green corridors may reduce local heat exposure, while dense built-up areas often retain heat more strongly.

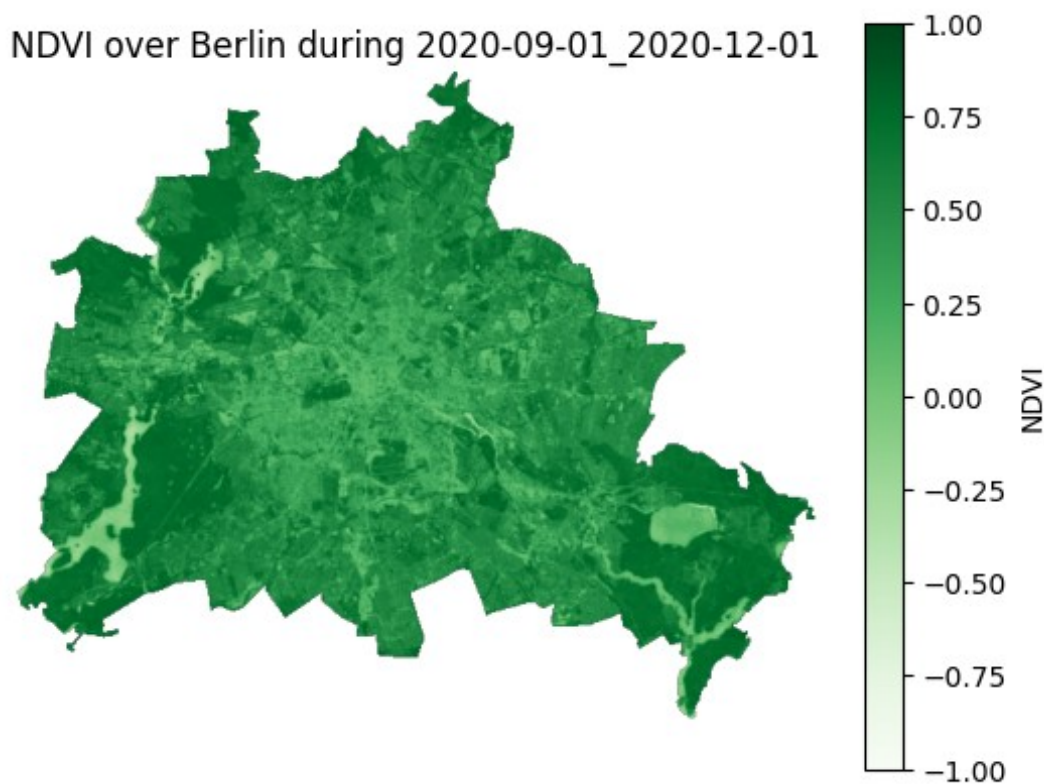


Figure 6. Berlin NDVI map for 2020-09-01 to 2020-12-01. Repository output reports mean NDVI = 0.509 for this quarter.

The notebook output reported a mean NDVI of 0.509 for Berlin during 2020-09-01 to 2020-12-01. This is a citywide average, so it should not be interpreted as the condition of every neighborhood. The map shows why citywide means need spatial context: Berlin contains both high-vegetation areas and lower-vegetation urban surfaces.

6.2 NDVI time-series pattern from 2020 to 2023

The quarterly mean NDVI time series shows a repeating seasonal structure. Values rise during greener periods and fall during winter or low-growth periods. For UHI analysis, this is important because vegetation is not static: the cooling potential of green areas changes across seasons. Any UHI model that uses NDVI should account for seasonal timing rather than using one annual average.

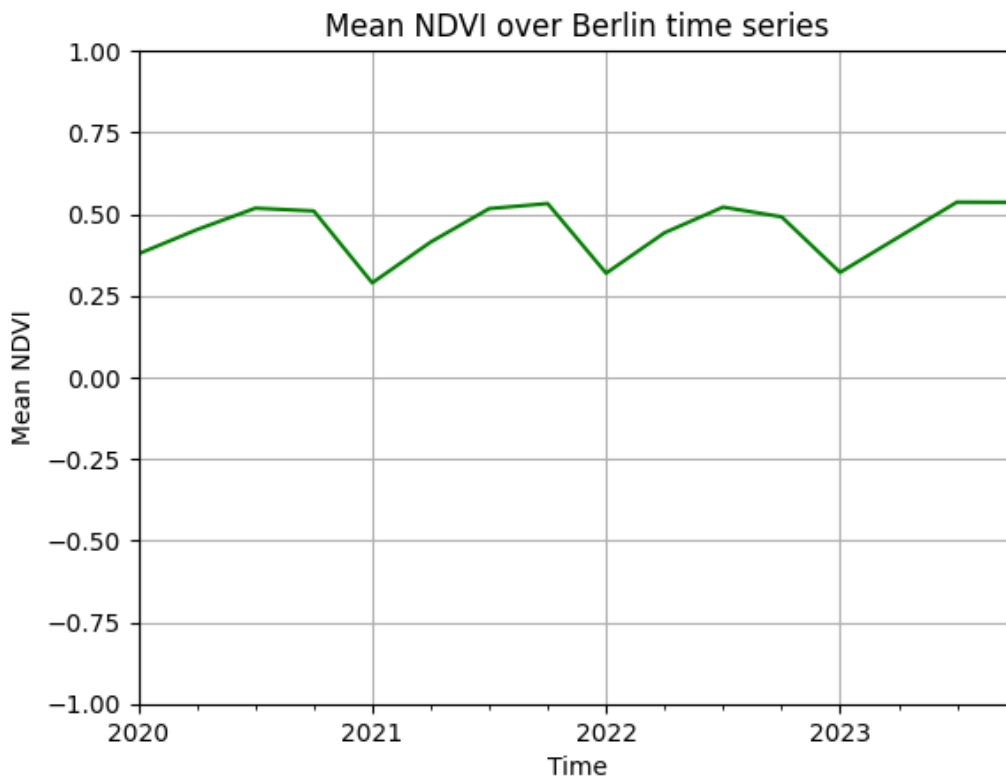


Figure 7. Quarterly mean NDVI over Berlin from 2020 to 2023, showing seasonal vegetation cycles.

The repeated pattern also creates a modeling opportunity. A future explanatory model could estimate whether the relationship between temperature anomaly and NDVI is strongest during warm-season quarters, when vegetation and heat exposure are both more relevant.

6.3 Ground-station temperature signal

The ECA&D station map shows that Berlin has enough within-city station coverage to support validation. The selected station, 4563, had 54,475 valid daily maximum-temperature records after applying the quality filter. The long time series confirms the expected annual temperature cycle, with summer peaks and winter lows. For a full UHI model, this station series should be compared with other Berlin stations and with rural reference stations outside the city boundary.

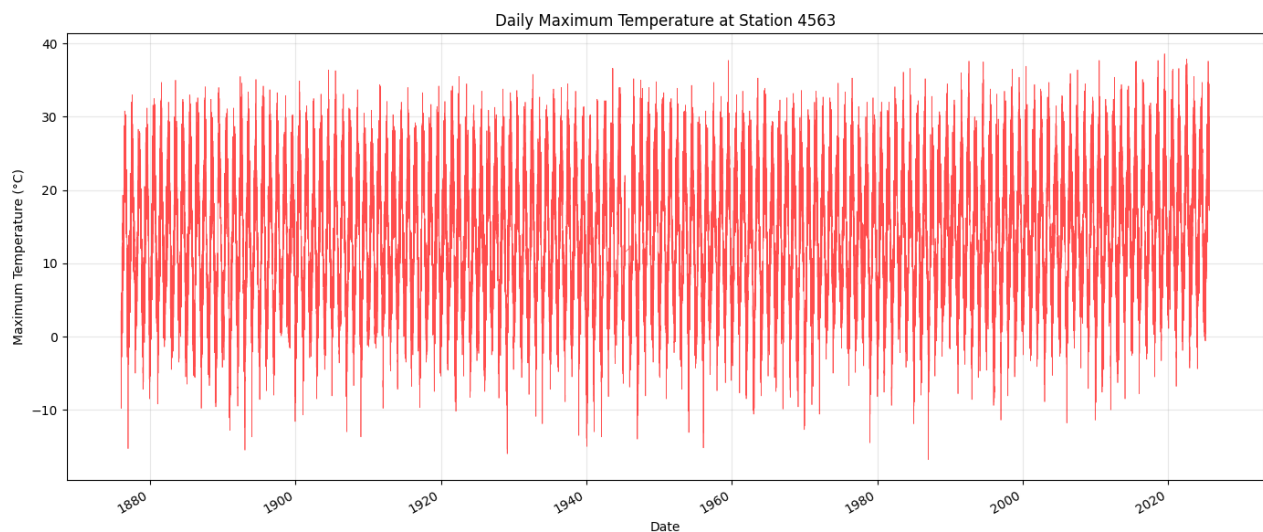


Figure 8. Daily maximum temperature at selected Berlin station 4563 after filtering valid ECA&D records.

6.4 What the visual results imply

Taken together, the repository outputs support three important analytical conclusions. First, the team successfully built a spatial integration pipeline: city boundary, NDVI raster, ERA5 grid, and weather stations were brought into a common analysis frame. Second, Berlin contains visible vegetation differences that are relevant to UHI interpretation. Third, the station data provide a path to validate satellite- and reanalysis-based findings rather than relying only on modeled climate products.

The project therefore should be presented as a data-integration and exploratory modeling project, not as a final causal estimate of UHI. Its strongest contribution is the reproducible framework that prepares the data needed for UHI measurement.

7. Code and Repository Quality

The repository includes multiple notebooks and saved exploratory reports. This is useful because it documents the progression from data exploration to city-specific analysis. It also reveals several improvement areas that can be discussed professionally in interviews.

- Strength: use of specialized geospatial libraries including GeoPandas, Rasterio, rioxarray, xarray, and regionmask.
- Strength: explicit data-quality steps, including nodata handling, quality-code filtering, CRS checks, and DMS coordinate conversion.
- Strength: saved figures and markdown reports for GADM and ECA&D exploration, making the project easier to review.
- Improvement: raw data paths are local and user-specific, so the project would benefit from config-based path handling and clearer instructions for replication.
- Improvement: the repository contains both Barcelona and Berlin exploratory branches. A portfolio version should clearly state Berlin as the final case study and move earlier city experiments to an appendix.
- Improvement: the final UHI metric should be implemented explicitly, for example by comparing low-NDVI urban zones against high-NDVI or rural reference zones.

8. Reflection, Limitations, and Future Work

8.1 Main limitations

- Data availability: raw datasets are not stored in the GitHub repository because the data folder is gitignored. This is reasonable for large files, but the README should describe where each dataset comes from and how to recreate the folder structure.
- City consistency: some notebooks and saved NDVI graphs refer to Barcelona, while the stronger final notebook uses Berlin. This should be cleaned for portfolio presentation.
- UHI metric not finalized: the existing work prepares the inputs but does not yet compute a full urban-minus-rural UHI intensity metric.
- Spatial resolution mismatch: ERA5-Land grid cells are coarser than Sentinel-2 NDVI rasters, so reprojected grids should be interpreted carefully.
- Station representativeness: one selected station is useful for exploration, but UHI measurement should compare multiple stations across land-cover classes and outside-city reference areas.

8.2 Recommended next steps

10. Create a clean portfolio branch focused only on Berlin.
11. Add a configuration file for all data paths and remove hardcoded local paths.
12. Classify Berlin pixels or districts into low-, medium-, and high-NDVI zones.
13. Use station observations to compute daily or seasonal urban heat anomalies.
14. Compare station-based anomalies with ERA5-Land estimates to quantify model discrepancy.
15. Fit an explanatory model such as temperature anomaly ~ NDVI + wind + precipitation + season.
16. Create a dashboard or final notebook with all figures, code, and interpretation in one place.

A simple but strong next model would be a seasonal regression that predicts temperature anomaly using NDVI, wind, precipitation, and month fixed effects. This would turn the exploratory work into a stronger applied data-science project while remaining interpretable for research-assistant or internship applications.

```
# Possible extension model structure
# temp_anomaly = station_temp - city_or_rural_reference_temp
model_features = ["mean_ndvi", "u10", "v10", "precipitation", "season"]
# Fit: temp_anomaly ~ mean_ndvi + wind + precipitation + season
```

9. How to Present This Project for Internships or RA Applications

This project should be positioned as a complete geospatial data-integration project. The most impressive points are not only the maps, but the fact that multiple sources with different formats were cleaned, standardized, and aligned.

Resume-ready version

- Modeled Urban Heat Island risk in Berlin by integrating multi-source climate, satellite, administrative-boundary, and ground-station datasets using Python.
- Processed Sentinel-2 NDVI rasters with Rasterio and GeoPandas, converting raw raster values into interpretable NDVI and masking data to the Berlin boundary.
- Aligned ERA5-Land climate grids, NDVI rasters, and ECA&D weather-station observations to support spatial comparison, visualization, and future UHI metric development.

Interview explanation

In an interview, I would explain this project as follows: I worked on a four-week climate analytics hackathon focused on the Urban Heat Island effect. My team selected Berlin because it had a clean administrative boundary, visible vegetation variation, and 20 weather stations within the city. I used Python tools such as GeoPandas, Rasterio, xarray, and rioxarray to process different data formats, convert NDVI values, handle missing data, reproject climate variables, and generate visualizations. The main challenge was not one model, but integrating messy real-world datasets into a consistent analytical workflow.

10. Conclusion

The GenHack project demonstrates a strong undergraduate-level data-science workflow: real-world problem framing, multi-source data integration, geospatial processing, visualization, and careful reflection on limitations. The Berlin case study shows how vegetation, gridded climate data, and station observations can be combined to study UHI risk. The project is already strong for internship and research-assistant applications because it shows technical tools, independent problem-solving, and

practical data thinking. The most important improvement is to implement a final UHI intensity metric and clean the repository into a single Berlin-focused reproducible pipeline.

Appendix A. Repository Files Used

- README.md: described Team 46 workspace, datasets, setup, and reproducibility notes.
- read-era5-netcdf_v2.ipynb: main Berlin-focused notebook covering GADM, ERA5-Land, Sentinel-2 NDVI, and ECA&D stations.
- NDVI/NDVI_Explore.ipynb and NDVI saved graphs: early NDVI exploration and vegetation time-series framework.
- week1/GADM_Allen.ipynb and GADM_Dataset_Exploration_Report.md: administrative-boundary exploration.
- week1/ECA_Blend_Allen.ipynb and ECA_Blend_Exploration_Report.md: weather-station metadata exploration.
- challenge instructions/brief_participants.pdf: official GenHack challenge requirements and evaluation structure.

Appendix B. Source Notes

This report is based on the uploaded repository ZIP and the GenHack 2025 participant brief. The raw climate and raster datasets are not included in the repository, consistent with the README note that raw datasets should not be pushed to GitHub. Figures were taken from saved repository images and extracted notebook outputs. Interpretations are limited to the outputs available in the repository and are framed as an exploratory analysis rather than a final causal estimate of UHI intensity.